

Human Elephant Conflict Task Force Workshop

Beijing, China, July 10-11th, 2009



Group Presentations at close of Workshop

Workshop Organizing Committee:

Ravi Corea – Coordinator & President, Sri Lanka Wildlife Conservation Society

Ajay Desai – Co-chair, IUCN/SSC Asian Elephant Specialist Group

Simon Hedges – Co-chair, IUCN/SSC Asian Elephant Specialist Group

Meenakshi (Mini) Nagendran, Program Officer, Asian Elephant Conservation Fund, U.S. Fish and Wildlife Service

Chandeep Corea – Operations Director and GIS Specialist, Sri Lanka Wildlife Conservation Society



IUCN-AsESG BASIC PRINCIPALS OF HUMAN-ELEPHANT CONFLICT MITIGATION

Working Group members:

Ajay, Peter, Simon, Pruthu, Vinod, Wahdi

1.

The AsESG recommends prioritizing conservation-based approaches for mitigation of Human-Elephant Conflict over animal rights based approaches (i.e. those approaches that prioritize individuals to the detriment of population survival). (Example of killing/removing a frequent raiding males to reduce animosity and therefore threats to whole population and its habitat.)

2.

In each HEC situation the AsESG recommends use of the mitigation technique that is the least detrimental to the elephant population.

3.

Need to be deal with the causes of HEC rather than the symptoms.

4.

We recommend that HEC mitigation is always conducted using an evidence-based/adaptive management approach (i.e. one based on monitoring of the effects of the method(s) on HEC rates and elephants).

5.

Addressing HEC requires a landscape-level approach.

6.

HEC mitigation is not only responsibility of the States' wildlife/conservation authorities (e.g. Forest or Wildlife Depts.) but also the private sector, donor agencies, banks, communities/individual farmers.

7.

There is no permanent solution to HEC, instead human-elephant coexistence requires long-term commitment to managing elephant populations, human landuse, and human behavior/attitudes.

8.

There is no single mitigation technique that will be applicable in all HEC situations.

GROUP I – ASSESSING HEC

Christy Williams

Peter Leimgruber

Vinod Kumar Yadav

Aswin Bangun

Mozaharul Islam

Donny Gunaryadi

Raymond Alfred

QuickTime™ and a
TIFF (Uncompressed) decompressor
are needed to see this picture.

contributing factors, intensity

Objective

Identifying suitable tools for assessment => match
HEC type & mitigation system

Issues

- National elephant policy
- Governance issues and development agenda
- Human population increase

Assess HEC

- Identify location where HEC occurs
- Assess/collect data
- Analyze data/situation/type HEC
- Match type of HEC with mitigation techniques

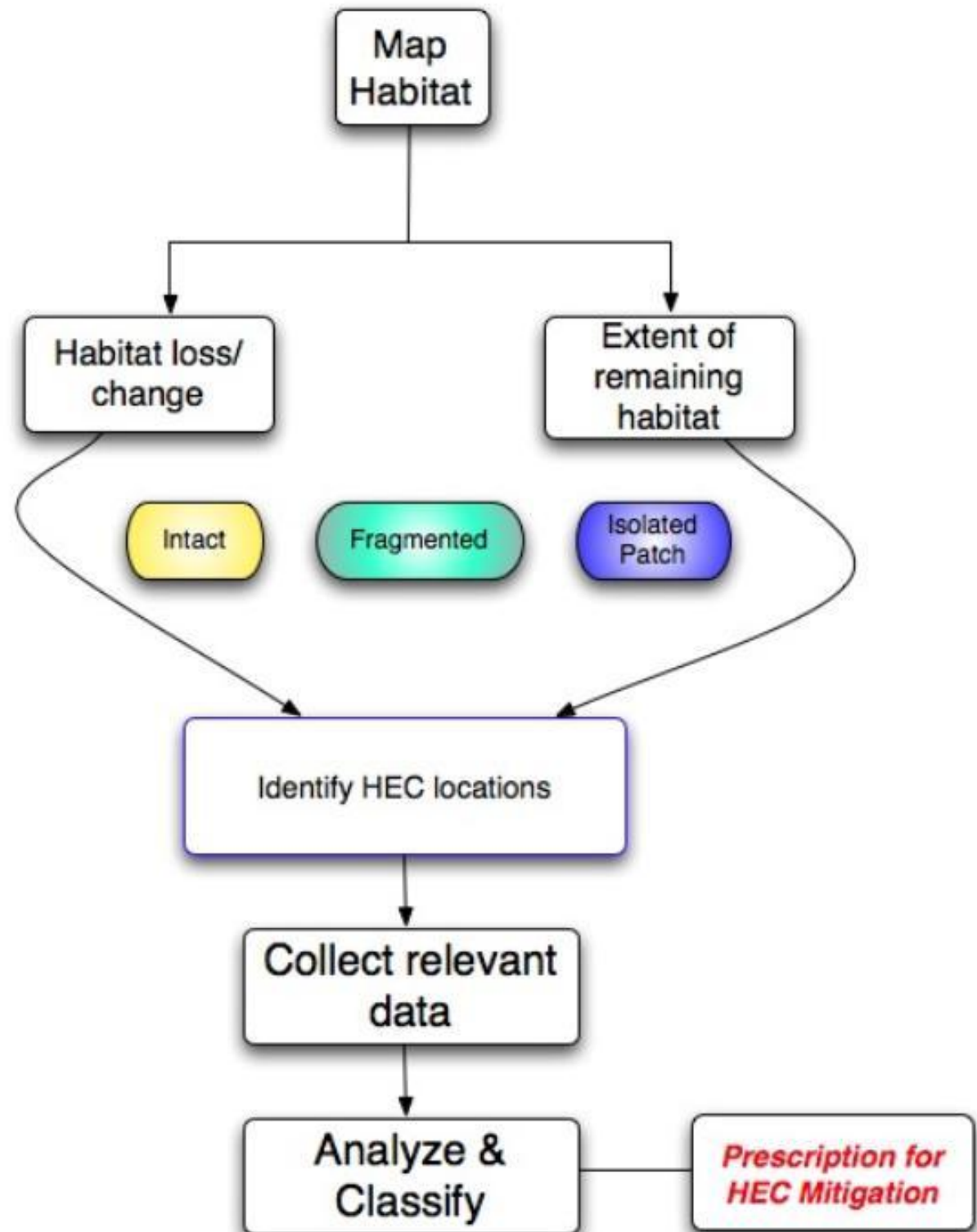
List of Data

Type of Data	Circumstance in which it is needed
Location (and forest cover land use at the location)	a. Basic data, needed for an initial assessment of whether HEC occurs. b. Habitat based classification (about extent, fragmentation and isolated populations)
Frequency of HEC incidents	- Elephant behavioural ecology based HEC incidents - Intensity of HEC
Crop/Property type Type & quantity of damage Area & quantity of damage Response questionnaires	1 Economic level of damage 2 People's tolerance level for the damage
Number of elephants involved Sex/age (raiding by tuskers) ID of elephants Radio Tracking	1. Need for Identifying the type of raiding (whether opportunistic raiders, habitual, obligatory, dispersing populations)

Decision Tree (Habitat)

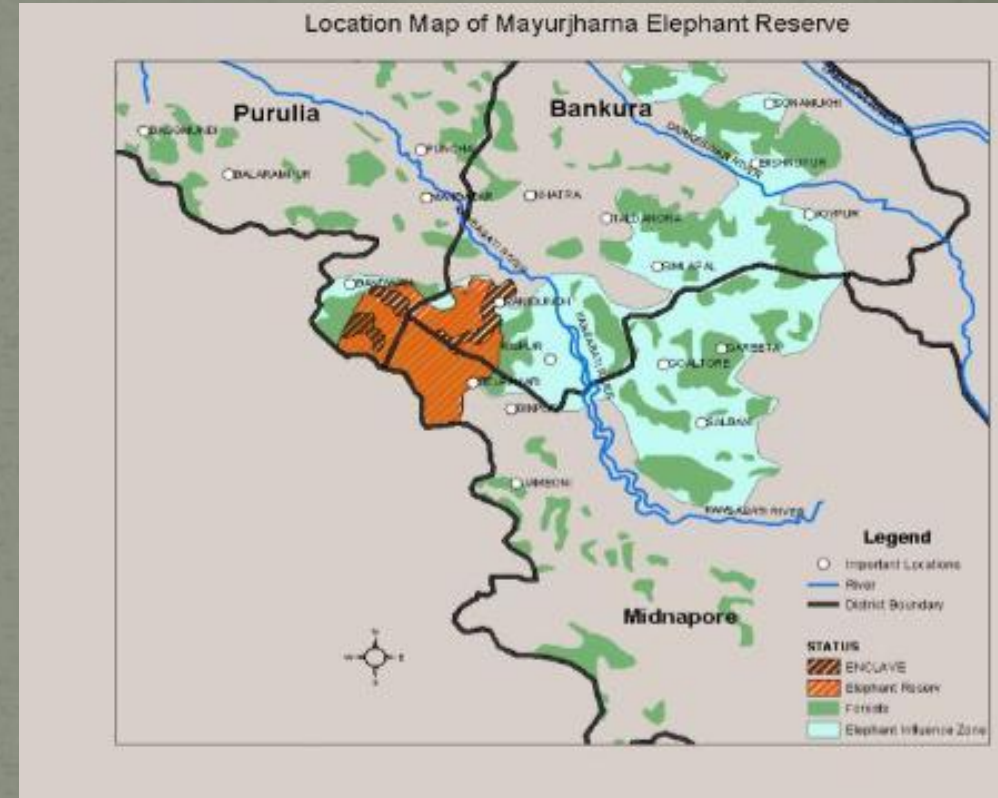
- Identify location
- Assess/collect data
- Analyze
data/situation/type
HEC

Classification system
=> match HEC type &
mitigation



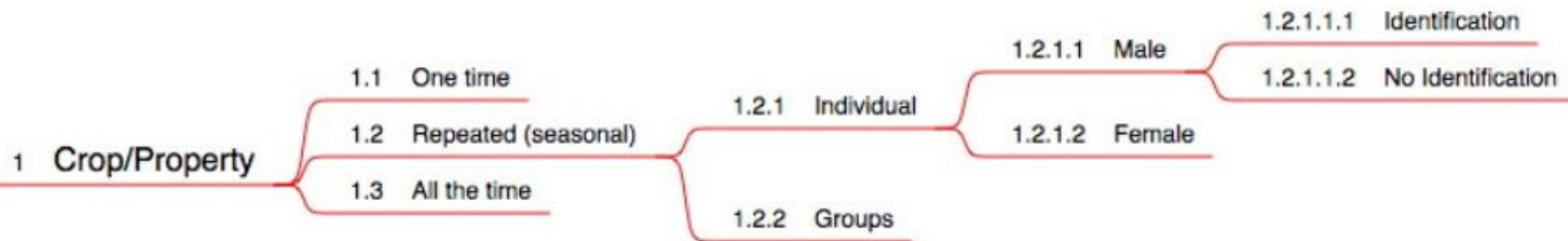
Methods

- Satellite images and GIS database
- Field surveys
- Location of hotspots (HEC and Forest loss)
 - Sampling based collection methods (refer to African Elephant Specialist Protocols, WCS HEC collection protocol)
 - GPS Location of ad-hoc reports and reports of conflict (as opposed to actual damage)
- Assessment of home range
 - Radio Tracking
 - GPS

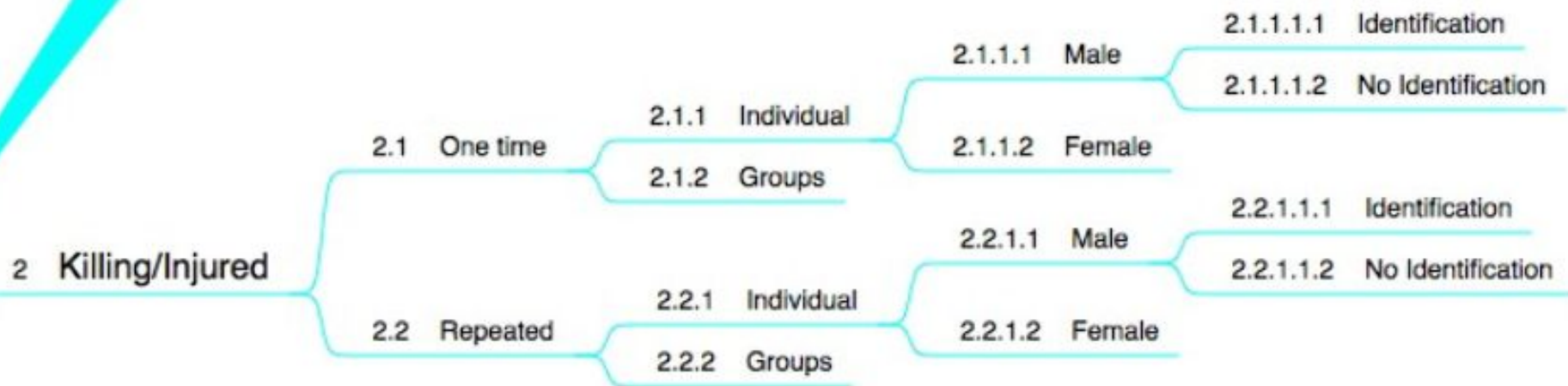


Decision Tree

(Assessing HEC Mitigation Needs)



HEC Assessment



Further Discussion

- Scaling the intensity of HEC (quantitative & subjective) - I.e. India - Sri Lanka & Sabah (Borneo) data

Low/Poor	Moderate/Fair	Good	Ideal/Very Good	Ultimate
0 - 0.3	0.3 - 0.8	0.8 - 1.5	1.5 - 2.0	Above 2.0
0 - 0.3 may indicates low viability, habitat may not suitable for the population and HEC may not happen in this area	0.3 - 0.8 indicates moderate viability (population), habitat may suitable, connected to the preferred forest habitat	0.8-1.5 indicates good viability (population), shows habitat is suitable and preferred forest habitat	1.5-2.0 indicates ideal viability (population), habitat is classified as key habitat	Above 2.0 indicates the habitat is highly used by elephant , which is due to forest conversion, limited availability of movement as corridor and forest fragmented HEC may or will occurs

(Sources: R. Alfred, Sabah (Borneo), 2009)

Classification of HEC types

- HABITAT AND POPULATION BASED CLASSIFICATION

- Adequate and reasonably intact (contiguous) habitat (>1000 km²; 1000-500 km²; <500 km²)
- Fragmented patches (<250 km²)
- Isolated habitat patch

- ELEPHANT BEHAVIOUR-ECOLOGY BASED CLASSIFICATION

- Opportunistic crop raiding
- Habitual crop raiding
- Obligatory crop raiding
- Dispersing populations (herds/bulls)

* ecosystem & management dependent

Guarding, patrolling, driving methods

- Guarding: ground huts, watch towers, noise (drum beating, shouting), information relay (singing, yodelling, cell phones), fire crackers, trip wires, light (fire torches, spot lights), chilli fence, live fences, direct use of domestic power, electric fence, trenches, village squads, dogs – need evidence for some and more data/information
- Patrolling: community/Forest-wildlife Dept. /CBOs- on foot, vehicles, elephants – more evidence and data needed
- Chasing/driving: noise based, fire based, chilli smoke, kunki (trained captive elephants), lethal methods (spears, arrows, burning rubber balls) - evidence and data needed
- Retaliatory killing: poison, shooting, new methods (explosive hidden in food, bed of nails) – illegal culling!! How do we respond to this?

Experiences of range countries

- Mitigation methods common across the range countries
- Guarding methods most popular but not organized
- Patrolling on foot common, vehicle and elephant use limited
- Day to day chasing common, long distance drives limited to case-to-case basis
- Traditional methods used extensively but new/introduced techniques confined to select locations as defined by the implementing agency (Line Dept, CBOs, etc.)
- What is also common is that there has been no systematic evaluation of the success/failure of these methods
- Often inappropriate methods are applied due to lack of resources or capacity

Success/failures, information gaps

- Methods used in combination likely to be more effective than “stand-alone” use
- Import to use low cost technology and locally available material where possible
- Level of desperation, economic status in the community influences acceptance of mitigation – based on our experiences
- Low levels of community commitment owing to different interest groups
- Need to measure the efficacy of various mitigation techniques
- Scope for innovation and reducing the cost of mitigation
- Need to gather more information (scientific papers, reports, etc.) to further evaluate these methods and identify information gaps
- Need to develop this section – more detailed information on techniques, costs, evaluation, etc.

Group 3-Barriers and deterrents

- Five range countries participated in the discussion on the efficacy of various barriers and deterrents.
- Barriers and deterrents can be used five situations-

Containing elephant movements

Protecting human habitation/property

Protecting crops/plantations

Containing isolated/pocketed population

Guiding the elephants/facilitating movement in corridor

- Barriers act as Physical/Psychological /Combined functions.
- Landscape buffers combined with forest and agriculture are more vulnerable to HEC.

Barriers- (more data needed)

- Seven categories of barriers have been used by various range countries, these are- Live and biological fences, wooden/wire fences, chilli and chemical fences, Power fences, trenches, metals/rail

- Among above power fence is relatively more effective as long as it is properly maintained with complete involvement of all stakeholders
- Other barriers like trenches and metal fence are effective but limited by their need of high cost, maintenance
- Any method that stops elephants accessing resources (crops or habitat) outside their main habitat can have some adverse impact on them. As such enclosing/excluding them must take this into account.
- Less effective barriers are chilli and other biological fences. But cheaper and easily available raw materials make them locally popular. Some of biological fences also provide economic returns.

Deterrents (more data needed)

- Five categories of deterrents have been used by various range countries, these are- chemical, light/fire, noise making device and odours.
- Elephants (especially in high conflict areas) but they get quickly habituated to some deterrents like light /fire and noise making devices. Altering the methods or in combination give better results.
- Scope of chemicals and odours as elephant deterrents are limited. Weather factors- wind & rain can make them less effective.
- Community involvement and participation is highly desirable components to make the deterrents more effective.

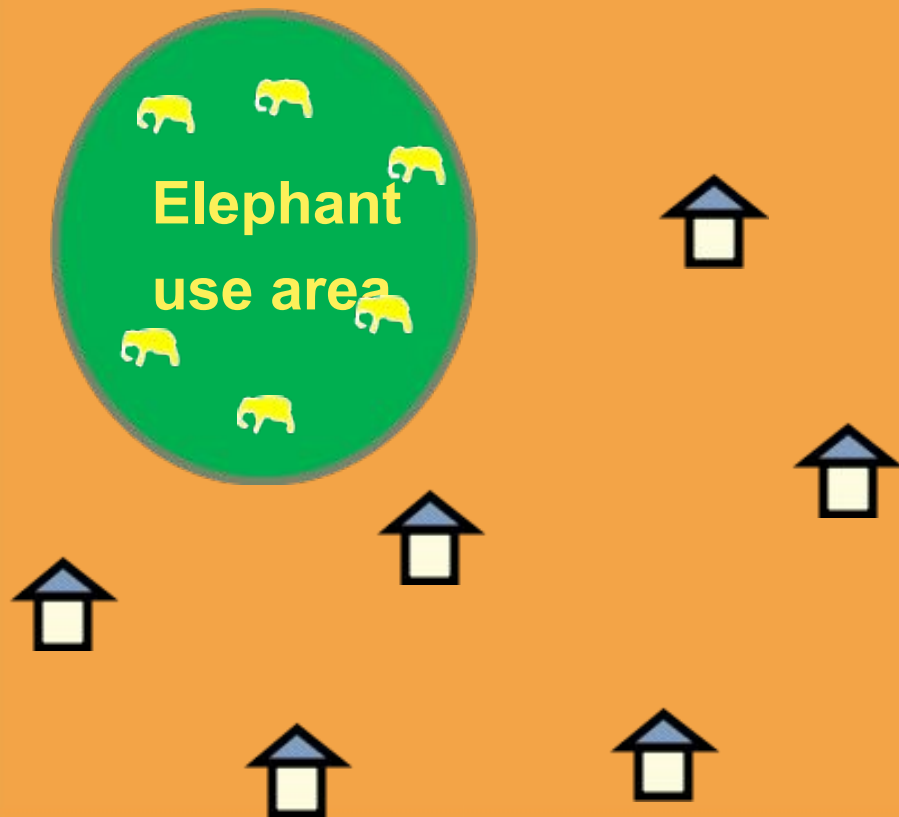
Options for Conflict Resolution at Human-Elephant Interface



Group 4

Elephants in Permanent Human Use Areas

Permanent Human use area



Most of the protected areas in
range countries

Permanent Human use area



Sri Lanka (DW), Kodagu
Karnataka, Maharashtra, India
(Prachi), Sabah, Southern China

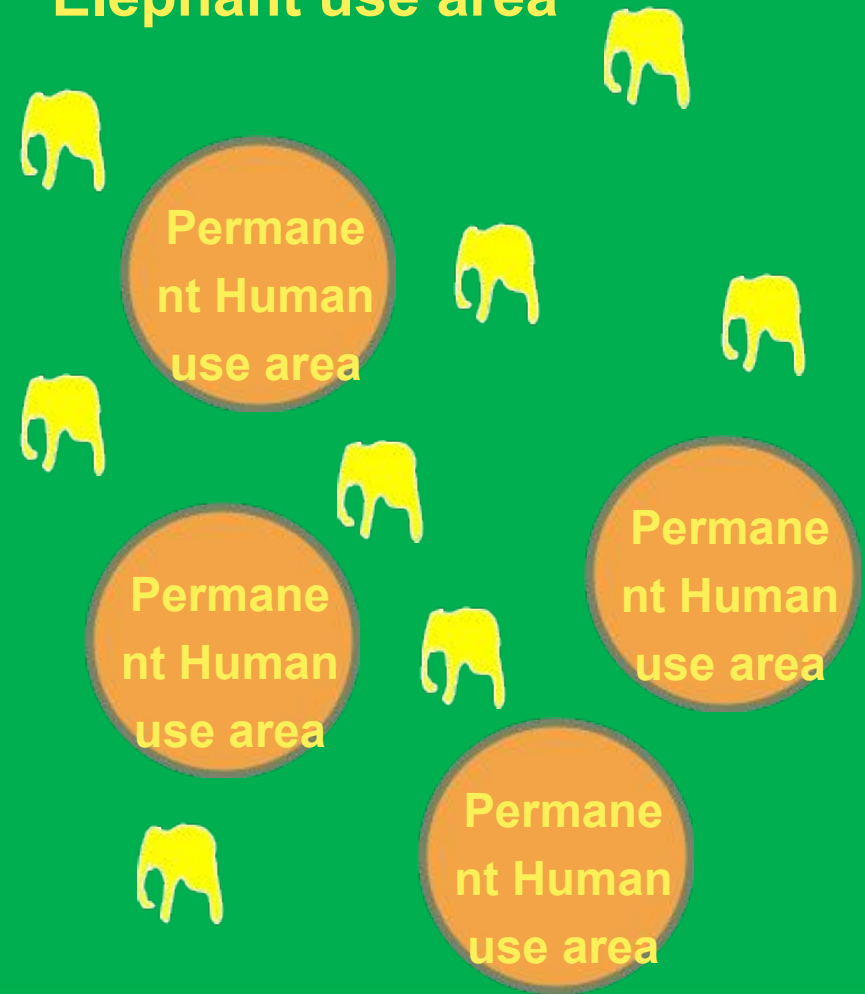
Permanent Human Use in Elephants Areas

Elephant use
area



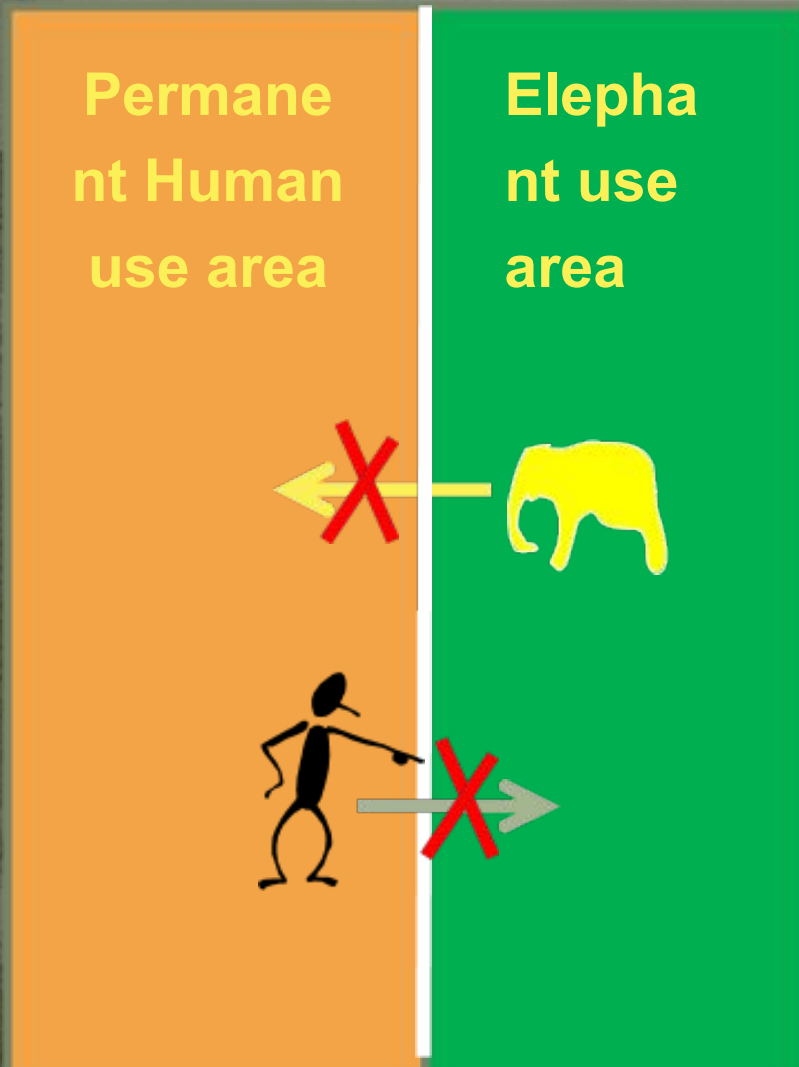
Thailand (Supol)

Elephant use area

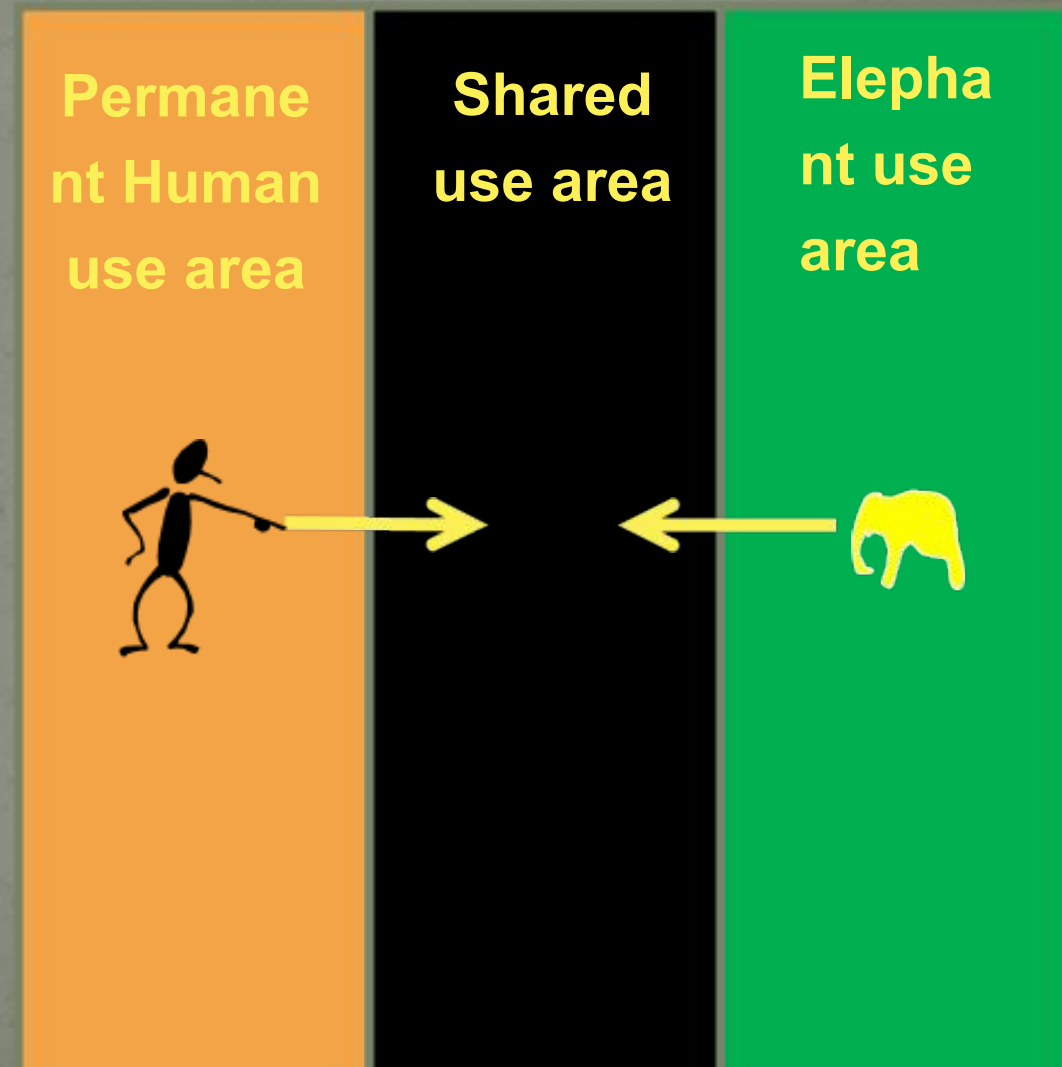


Indonesia (Donny), Tal India (KD), Assam, Northeast India, Cambodia, Sri Lanka, Lao

Elephant- Permanent Human use interface



Elephant - Human shared use interface



Elephants Surrounded by Permanent Human Use Areas

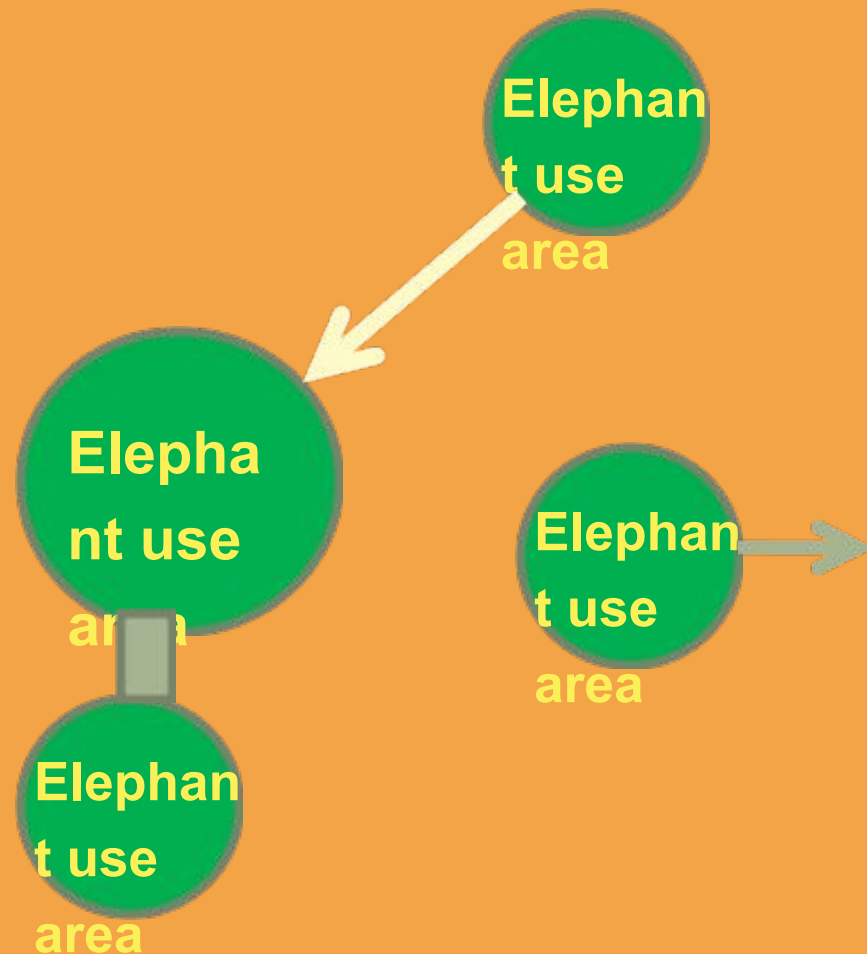
Permanent Human use area

Elephant use area

- Population Viability Assessment:
e.g. Numbers, genetic diversity, habitat quality etc., proxy:
Available habitat
- Viable
 - Hard boundary
 - Barrier
 - Unpalatable food plants (Sri Lanka, India - Nanditha)
 - Livestock (Thailand - Simon)
 - Alternatives for forest dependencies (KD - Thailand)
 - Shared use boundary
 - Tourism (Lao - Alex, Sabah - Isabelle)
 - Temporary farming with temporary barriers (Sri Lanka, Thailand)

Elephants Scattered in a Permanent Human Use Area

Permanent Human use area



- Population Viability Assessment: e.g. Numbers, genetic diversity, habitat quality etc.,

If viable

- Barriers (see previous slide)
- Corridors to link patches where elephant movement occurs and move people from corridor area

If non viable:

- Translocate to viable patches
- Removal of elephants from the area

Permanent Human Use Surrounded by Elephant Use Area

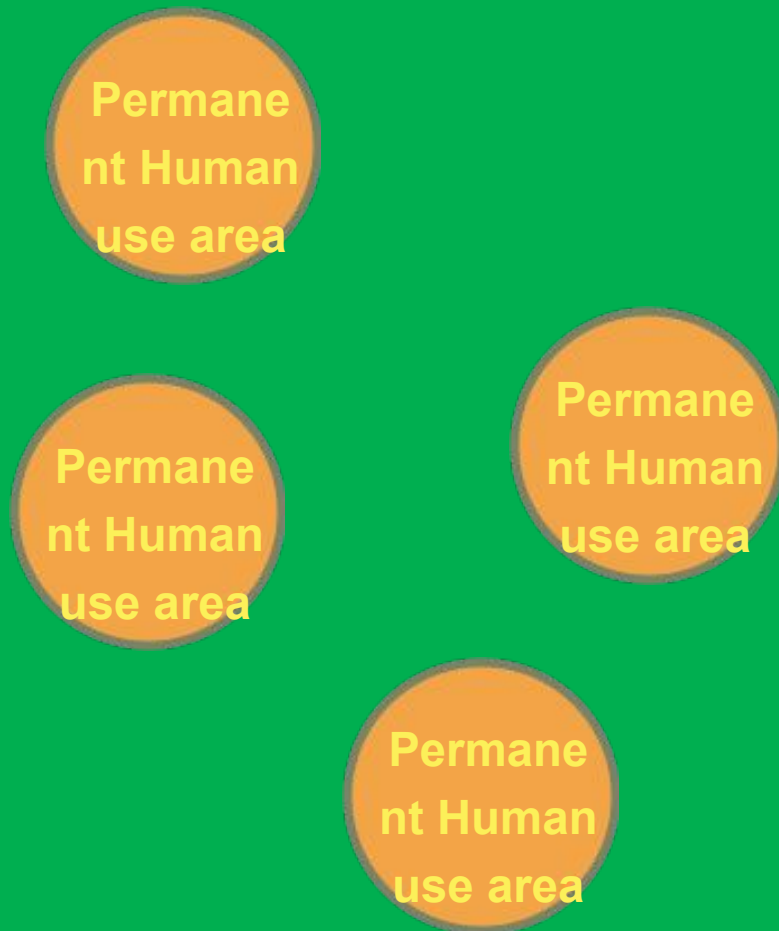
**Elephant use
area**

**Permane
nt Human
use area**

- Removal of the village
- Permanent barrier around the village
- Land use changes within villages to make it less attractive to elephants
- Promote elephant based tourism

Scattered Human Use in Elephant Area

Elephant use area



- Cluster human use areas together
 - Drive elephants out of human use area and establish a hard or shared use area
-

Thank You



Group 6 - Monitor and measuring human elephant conflict mitigation success/failure

Group members:

Pruthu Fernando (Sri Lanka)

Ananda Kumar (India)

Rinjan Shrestha (Nepal)

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Approach

Objectives

Methods/Activities

Effectiveness assessment

Specific objectives

Impact on HEC

Impact on elephants

Variables

Tools



COMPENSATION FOR LOSS FROM HUMAN ELEPHANT CONFLICT

Part 1. Direct Payments

1.1 Cash payments

- Status of cash payment schemes in range countries* [e.g. Malaysia-very limited, some compensation for crop loss in Peninsula but none in Sabah]
- Pros and cons
- Management agencies [e.g. government, NGO, community-based]
- Sources of funds [industry, tourism, etc]
- Monitoring requirements
- Further actions needed [clear guidelines, recommendations]

1.2 Insurance

- Status of insurance schemes in range countries* [e.g., Bangladesh - no insurance scheme for HEC but exists for tiger-human conflict]
- Pros and cons
- Management agencies and sources of funds [could be managed / funded by government, private sector/I-NGOs, community or any combination of these]
- Further actions needed [still a novel idea, requires further investigation]

* *Document existing policies / schemes*

COMPENSATION FOR LOSS FROM HUMAN ELEPHANT CONFLICT

Part 2. Indirect Payments

Techniques being trialed or considered and precautions to be taken

2.1 Alternative income

- Agro-forestry [e.g., Sri Lanka-reforestation with cash crops]
- Cultural / handicraft [several examples but lack evidence of success]
- Ecotourism [Nepal-tourism income supports HEC mitigation compensation]
- Non-timber forest products

2.2 Improving income from existing sources

- Agriculture [e.g. Sri Lanka-fencing to improve crop production]
- Improved marketing [e.g. Elephant Pepper Development Trust]
- Value added [Sumatra –incentives to relocate coffee outside PA]
- Micro-finance [several examples from Sumatra]

2.3 Livelihood improvement linked to reducing HEC

- Education and health [trials in Nepal underway]

COMPENSATION FOR LOSS FROM HUMAN ELEPHANT CONFLICT

Steps needed to complete this chapter

- Chapter existing schemes and their evaluation; add associated references
- More review and contributions from other multi-disciplinary practitioners

Steps needed to address the practice of compensation

- Compensation for HEC loss is a complex issue
- Evaluation of compensation schemes to determine change in individual and/or community attitudes / behavior towards HEC and towards conservation of elephants and their habitats
- Are there compensation schemes that could be applicable across all range states?

Human

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